

Experiment 4: Packet Arrival Stream Processing

Aim

To simulate a network packet arrival stream by generating packets of random sizes one after another, processing each packet as it arrives, and visualizing the packet arrival stream using a line plot.

Theory

In a computer network, data is transmitted in small units called packets. Packets do not arrive all at once, they arrive one after another as a continuous stream. The size of each packet is not fixed and normally varies between 64 bytes and 1500 bytes, where 1500 bytes is the maximum transmission unit of an Ethernet frame. Processing such packets as they arrive, instead of waiting for the entire data to be collected, is an example of stream processing. Plotting the packet sizes in the order of their arrival helps in identifying traffic patterns and bursts in the network.

Requirements

- Python 3.x
- Jupyter Notebook
- Random module
- Time module
- Matplotlib

Step 1: Import Libraries

```
import random
import time
import matplotlib.pyplot as plt
```

Interpretation: Imports the modules required to generate random packet sizes, simulate the arrival delay and plot the stream.

Step 2: Generate Packet Stream

```
packet_stream = []

for i in range(20):
    packet_size = random.randint(64, 1500)
    packet_stream.append(packet_size)
    print(f"Packet {i + 1} arrived: {packet_size} bytes")
    time.sleep(0.2)
```

Interpretation: Generates 20 packets with random sizes between 64 and 1500 bytes. Each packet is stored in the stream and printed as soon as it arrives.

Step 3: Visualize Packet Arrival

```
plt.plot(packet_stream, marker="o")
plt.title("Network Packet Arrival Stream")
plt.xlabel("Packet Number")
plt.ylabel("Packet Size (Bytes)")
plt.grid(True)
plt.show()
```

Interpretation: Displays the packet arrival stream as a line plot with markers, showing how the packet size changes from one packet to the next.

Complete Program

```
import random
import time
import matplotlib.pyplot as plt

packet_stream = []

for i in range(20):
    packet_size = random.randint(64, 1500)
    packet_stream.append(packet_size)
    print(f"Packet {i + 1} arrived: {packet_size} bytes")
    time.sleep(0.2)

plt.plot(packet_stream, marker="o")
plt.title("Network Packet Arrival Stream")
plt.xlabel("Packet Number")
plt.ylabel("Packet Size (Bytes)")
plt.grid(True)
plt.show()
```

Sample Input

No external input file is required. The packet stream is generated at runtime:

```
Number of packets      : 20
Minimum packet size    : 64 bytes
Maximum packet size    : 1500 bytes
Delay between arrivals : 0.2 seconds
```

Expected Output

```
Packet 1 arrived: 727 bytes
Packet 2 arrived: 372 bytes
Packet 3 arrived: 872 bytes
Packet 4 arrived: 1397 bytes
Packet 5 arrived: 162 bytes
...
Packet 20 arrived: 1094 bytes
```

A line plot of packet size against packet number is then displayed with circular markers. The exact values change on every run because the sizes are generated randomly.

Interpretation

- Each packet arrives one after another.
- Packet size varies from 64 to 1500 bytes.
- Simulates real network traffic.
- Displays packet size over time.
- Helps identify traffic patterns and bursts.

Result

The network packet arrival stream was successfully simulated and visualized. Twenty packets of random sizes were generated, printed as they arrived, and plotted to show how packet size varies

over time in a network.